Ruthenium-106 Brachytherapy and Central Uveal Melanoma

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Abstract

Purpose

Uveal melanoma (UM) is the most common primary ocular malignancy. The size of the tumor and its location are decisive for brachytherapy with β-emitting Ruthenium-106 (Ru-106) plaque. The treatment of juxtapapillary and juxtafoveolar UM may be challenging due to the proximity or involvement of the macula and optic nerve. High recurrence rates have been observed.

Methods

Central UM was defined as a lesion up to 5mm off the optic disc or fovea- radius of 5mm. Between January 2011 and July 2020, we treated 56 patients with Ru-106-brachytherapy. The clinical outcome for recurrence, visual acuity, and radiation-related toxicity was assessed. The follow-up was 66 (6-136) months.

Results

Of the 56 patients (56 eyes), eight patients (14%) suffered from local recurrence. Six relapsing UM of 19 patients (32%) were located close to the optic disc, and two patients had UM close to the macula (2/37, 5%), ($p > 0,05$). The overall rate of eye preservation was 89%. Visual acuity (VA) before treatments was 0,45 and was reduced to 0,26 after brachytherapy. Radiation retinopathy or opticopathy was detected in seven patients (13%), and radiation maculopathy in ten patients (17,9%). Six patients (11%) underwent enucleation due to recurrence or radiation-induced ophthalmopathy.

Conclusion

Therapy of central UM is challenging. We suggest, that central UM should be categorized as lesions laterally or medially to the fovea due to different likelihood of long-term control rates. Localization near the optic disc requires thoughtful management.

Introduction

Uveal melanoma (UM) is the most common primary ocular malignancy with an incidence in Europe of 1:100,000 per year [18]. Prognosis is determined by histology, genetic profile, the base diameter, prominence, location and ciliary body involvement. Every millimeter increase in apical height leads to a 5% increased risk for metastasis [20]. Distant metastases occur in up to 25% of patients after five years [6, 22].

Depending on the size of the tumor and its location in relation to the optic nerve, the macula and the ciliary body, different treatment options are available. In addition to brachytherapy in the form of contact therapy with Ruthenium- 106 (Ru-106) or Iodine- 125 (I-125), external beam radiation therapy options include proton beam therapy and stereotactic radiotherapy with photons [5]. Brachytherapy is preferentially used for peripheral UM with an apical height up to 6 mm or central UM close to the fovea with an apical height up to 6 mm. We use stereotactic radiotherapy with photons successfully for UM with an apical height more than 6mm and UM close to or abutting the optic disc and achieved high local control rates [24]. Regarding the personal situation of each patient such as high age, underlying diseases and immobilization different treatment options for a personalized therapy such as Ru-106 Brachytherapy with adjuvant transpupillary thermotherapy is also a treatment possibility.

Brachytherapy with β-emitting Ru-106 plaque is an effective and widely used treatment option first introduced by Lommatzsch and Vollmar [12]. The beta rays from the radiation shield ensure easy handling and optimal tumor dose coverage limited by the height of the UM for Ru-106 [7, 25].

UM close to the macula and optic nerve is challenging, because the central presentation is a risk factor for local treatment failure. Precision of the placement of the plaque is key to minimize the risk of damage of the central sensitive structures through radiation complications for example radiation retinopathy and optic neuropathy, toxic tumor syndrome and cataract and maintain a good visual acuity (VA).

Since January 2011, we have established Ru-106-brachytherapy as a treatment option in our hospital. Here we present our results of Ru-106-brachytherapy for successful therapy of central UM.

Patients and Methods

From January 2011 to December 2019 a total number of 283 of UM were referred to our institution. Ru-106- brachytherapy was used in 171 patients, three eyes were enucleated and fractionated stereotactic radiosurgery was used in 109 patients.

In this retrospective cohort study, patients treated with Ru- 106- brachytherapy and a localization of the UM touching the macula were included. A total of 56 eyes of 56 patients with central UM were analyzed. Central UM were defined as a juxtafoveolar UM with a posterior edge abutting the optic disc- fovea- radius around the fovea by five mm. The cases were divided in two groups. Group I were UM reaching the nasal half of the five mm discoid area around the fovea:19 eyes (34%). Group II consisted of patients with an UM localized on the temporal half of the foveal disc of five mm. (37 eyes, 66%). If an UM touched both sides of the fovea, it was scored as group I (Fig. 1).

The diagnosis was made based on the clinical findings, using sonography A- and B-scan. Additionally, in a minority of cases the diagnosis was confirmed using fluorescence angiography or magnetic resonance imaging studies. Criteria for malignancy were orange pigment, exudative changes such as an accompanying detachment or a documented tumor growth.

In nine cases, a Ru-106 plaque CCA with a diameter of 15,3 mm, in 16 cases a Ru-106 plaque CCB with a diameter of 20,2 mm and in 31 cases a Ru-106 plaque CCD with a diameter of 17,9 mm was used (Eckert & Ziegler, BEBIG®, Berlin, Germany). The plaque was fixed onto the sclera with supramid 6 – 0 and the conjunctiva was closed with 7 – 0 vicryl in general anesthesia. The plaque was left in situ for a specific time. A dosage of about 110 Gy at the apex of the tumor was calculated. The plaque was removed in general anesthesia.

The mean apical height was 3,9 (0,9 – 8,28) mm, in group I 4,39 mm and in group II 3,67 mm. Nine eyes showed an apical height more than 6mm. 36 UM (64%) had an exudative retinal detachment prior to brachytherapy, 15 in group I (79%) and 21 in group II (57%). We used a CCD plaque in 31, a CCB plaque in 16 and a CCA plaque in nine cases. The mean scleral dosage was 480 (200–1000) Gy and the mean dosage at the apex 105 (84–105) Gy.

Patients were reviewed 3 months, 9 months and then yearly after brachytherapy. At each follow- up visual acuity, clinical examination, sonography, fundus photographs were performed. Metastasis surveillance by liver ultrasound was carried out 6 monthly.

After obtaining written informed consent from the patients, the researchers conducted the clinical examinations which are described below. Along with the approval of the Investigational Review Board from local ethics committee, the study adhered to the tenets of the Declaration of Helsinki.

Statistical analysis was performed with MATLAB® & SIMULINK®, Version R2007a (The MathWorks, Inc., Natick, MA, USA). For categorical variables, number (n) and percentage (%) are presented. For continuous variables, mean, and range are provided. For comparisons between groups the Mann–Whitney U test was used for continuous variables. A p value < 0.05 was considered statistically significant.

RESULTS

In this retrospective case series, we included 56 eyes of 56 patients (29 female, 27 male) with central UM, 24 (42%) right eyes and 32 (58%) left eyes. The mean age was 65 (40–89) years. The mean follow- up time was 66 (6- 136) months. The mean presenting decimal visual acuity (VA) was 0,45 (group I VA 0,37, group II VA 0,49) and the final VA 0,26 (group I VA 0,15, group II VA 0,32, p = 0,013) (Fig. 2).

Eight eyes (14,3%) showed a recurrence during the follow- up period. In group I there were six eyes (31,6%), in group II were two eyes (5,4%).

Overall, we observed a local tumor control rate of 85,7% (48/56), in group I 68,4% (13/19) and in group II 94,6% (35/37) (Fig. 3). After the relapse five of these eight eyes were additionally treated with fractionated stereotactic radiosurgery and three eyes were enucleated. Recurrences were observed after five to 49 months, the median time was 38,2 months. Three patients had their treated eye enucleated due to high ocular pressure or hemorrhage. The eye preservation rate was 89%. Group I showed an eye preservation rate of 4/19, 79%, and group II 2/37, 95%.

Distant metastasis was observed in seven cases (group I: 3, group II: 3) 14 to 100 months after brachytherapy. Nine patients died during follow-up (overall survival 83,9%), and six died of distant disease progression resulting in a cancer-specific survival rate of 89,3%.

Endoresection as a post-therapeutic intervention was necessary in four patients (group I: 1, group II: 3). Vitrectomy with silicon oil placement was done in five eyes (group I: 2, group II: 3) due to persisting exudative retinal detachment after brachytherapy.

Radiation retinopathy was detected in seven eyes (group I 2/19, 11%, group II 5/37, 14%), a radiation opticopathy in seven eyes (group I 5/19, 26%, group II 2/37, 5%) and a radiation maculopathy in 10 eyes (group I 2/19, 11%, group II 8/37, 21,6%) and six eyes (group I: 1, group II: 5) showed a macular edema. The macular edema was treated in three cases with intravitreal steroids and in one case with topical non steroid anti- inflammatory eye drops (Nevanac®).

DISCUSSION

In this retrospective study we showed that ruthenium plaque brachytherapy is a suitable treatment for central UM. The location laterally or medially to fovea impacts on cure rates. Tumor size and localization are established risk factors for local control. Due to the limited penetration power of ruthenium the treatment of melanomas thicker then 6mm can be insufficient. A localization close to the optic disc or center of the fovea is associated with a higher recurrence rate [2]. Technically, placing the plaque in a central position can be challenging. On the one hand you want to preserve as much VA as possible therefore you are trying to have the central border of the plaque close to melanoma and as less as possible close to the fovea (eccentric placement) and on the other hand you have the optic disc as an anatomical limitation [16] (Fig. 3). Our aim was to investigate the role of the localization of the UM to the optic disc, and to score for recurrences and complications.

The meta- analysis by Karimi *et al.* including 21 peer- reviewed studies involving 3913 patients with UM showed similar results regarding the local tumor control rate of 84% ranged from 59%- 98% [10]. O´Day *et al.* reported a treatment failure in their study with posterior UM of 15,5% (34/219) [14], similar as reported in the present series. We achieved an overall local tumor control rate of 85,7%. Tumor location was the only reason for the difference in control rate of 69,4% in group I and 94,6% in group II. Thickness was not an exclusion criterion, and nine eyes with tumors larger than 6mm were included. The reason for treating patients with lesions exceeding 6mm was due to patient´s preference, as the two of the nine patients with UM larger than 7mm were 87 and 80-years old, respectively, and refused primary enucleation or external beam radiation therapy, insisting to be treated with brachytherapy. Both remained without disease progression for 35 and 23 months, respectively.

The risk of recurrence in central UM varies in different studies between 69%-100% and a decreasing distance of the tumor margin to the optic disc up to less than 2 mm showed a higher recurrence rate [8, 9, 13, 16, 23].

Of eight eyes with a recurrence, five were treated with salvage fractionated stereotactic radiosurgery, and three eyes were enucleated. (3 + 5 sind 8: ok). There was no difference in the apex and scleral contact dosage regarding recurrence, localization, enucleation. Three eyes had to be enucleated because of the recurrence and three due to other circumstances such as high intraocular pressure and hemorrhage as a radiation complication. Our rate of preserving the globe was 89% overall. Likely to other studies globe retention was 95,4% [14], 84% at five years in juxapapillary UM [17], 89% for UM with macular involvement [9]. External beam radiation therapy such as proton beam or stereotactic radiosurgery have shown in several studies enucleation rates between 8% an 16% similar to those with brachytherapy [1, 11, 19].

Radiation complications such as retinopathy was detected in seven (12,5%) eyes, 11% in group 1, and 14% in group II 12,5%, radiation opticopathy in seven (12,5%) eyes, five (26%) in group I, and two (5%) in group II, and radiation maculopathy was reported in 10 (17,9%) eyes, two (11%) in group I, and eight (21,6%) in group II. Depending on the tumor hight and the distance to the optic disc and fovea radiation complication varies. Russo *et al.* found a radiation maculopathy in 7,4%, opticopathy in 1,8%, Sagoo *et al.* reported a radiation retinopathy non proliferative/proliferative in 66%/24%, a radiation opticopathy in 61% an maculopathy in 56% after five years [16, 17]. The therapy of radiation complication contains laser coagulation, off- label intravitreal Anti- VEGF and off- label intravitreal corticosteroids [26].

Radiation complications implements often a loss of VA. In addition several studies have shown that the treatment of central tumors are associated with a visual loss [4, 9, 15–17]. The mean presenting decimal visual acuity (VA) in our study was 0,45 (group I VA 0,37, group II VA 0,49) and the final VA 0,26 (group I VA 0,15, group II VA 0,32). Shields et al showed that central UM less than 5mm from the optic disc or foveola have in 35% or 25% after five years poor VA 0,01 or less up to no light perception [21].

In addition to brachytherapy, central UM can also be treated with proton beam or fractionated stereotactic radiotherapy. The first study that compared both treatment modalities showed that there is no difference regarding the local control and adverse side effects. Vitreous hemorrhage may be more likely in the group treated with stereotactic radiotherapy. Concerning the radiation maculopathy and opticopathy in central UM there was also no difference [3]. The eccentric plaque placement in brachytherapy offers an option to spare out the fovea in the treatment and maintain a good VA. This can be promptly controlled through direct binocular fundoscopy while placing the plaque. Concerning all treatment options an early detection of the UM and fast treatment is necessary for effective therapy because the size of the UM has an impact of the treatment success [3, 24].

In conclusion, despite the changes of central UM, our study shows that central UM can be treated well with Ru-106 plaques providing good rates of local tumor control of 86%. Due to the different likelihood of long-term control rates, central UM should be classifed as lesions laterally or medially to the fovea. Especially the localization near the optic disc requires thoughtful management and other therapeutic options such as external beam radiation therapy may be preferrable.

Declarations

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The authors have no relevant financial or non-financial interests to disclose.

All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by L. Grajewski, M. Wösle. The first draft of the manuscript was written by L.Grajewski and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

This is an observational retrospective study. Therefore no ethical approval is required in Germany.

Informed consent was obtained from all individual participants included in the study.

The authors affirm that human research participants provided informed consent for publication of the images in Figures 1, 3A and 3B."

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Figures

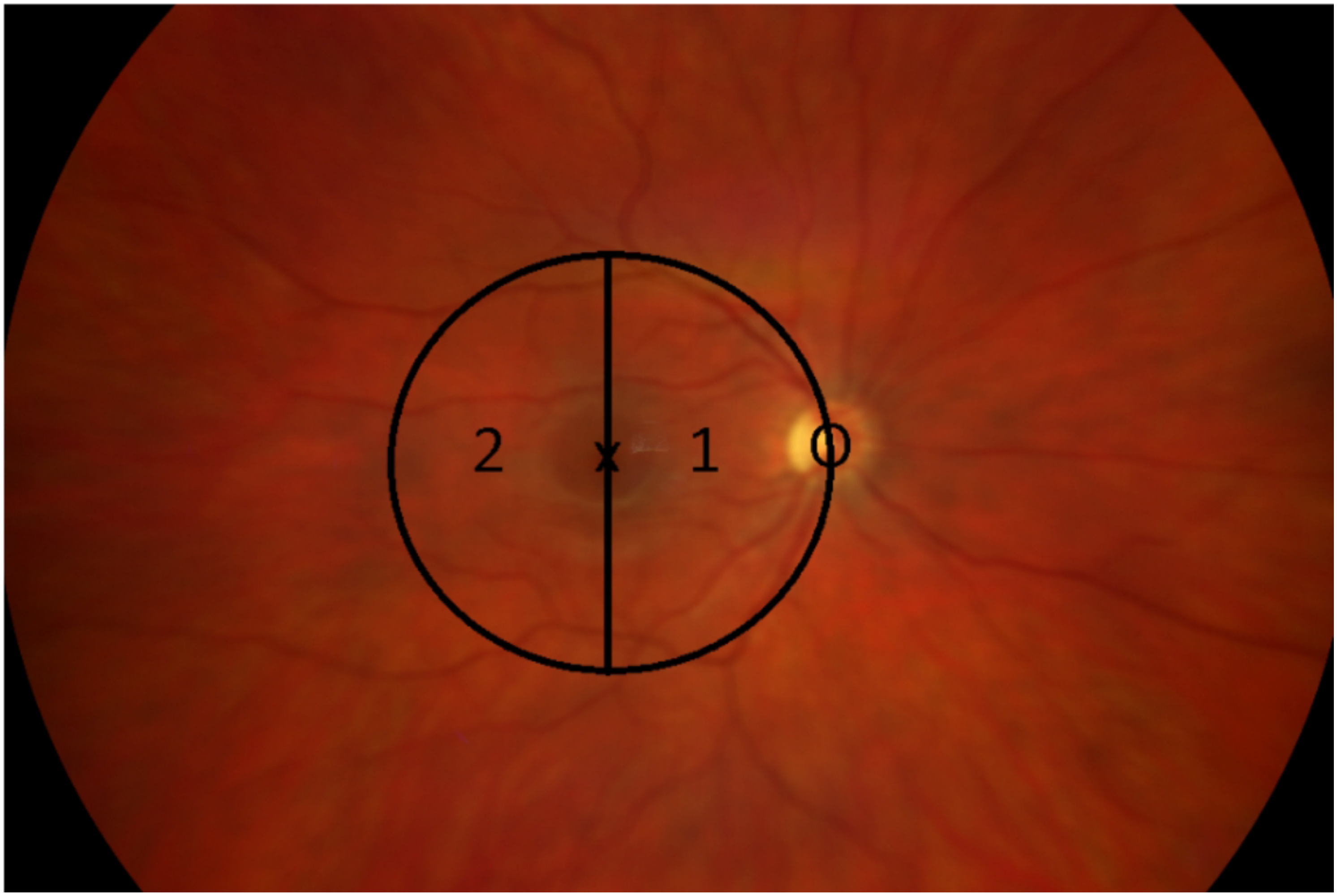


Figure 1

Categorization of tumor presentation according to the location as a function of the relation to the fovea. The fundus of a right eye: 1: group I; 2: group II; X: Fovea; O: Optic disc

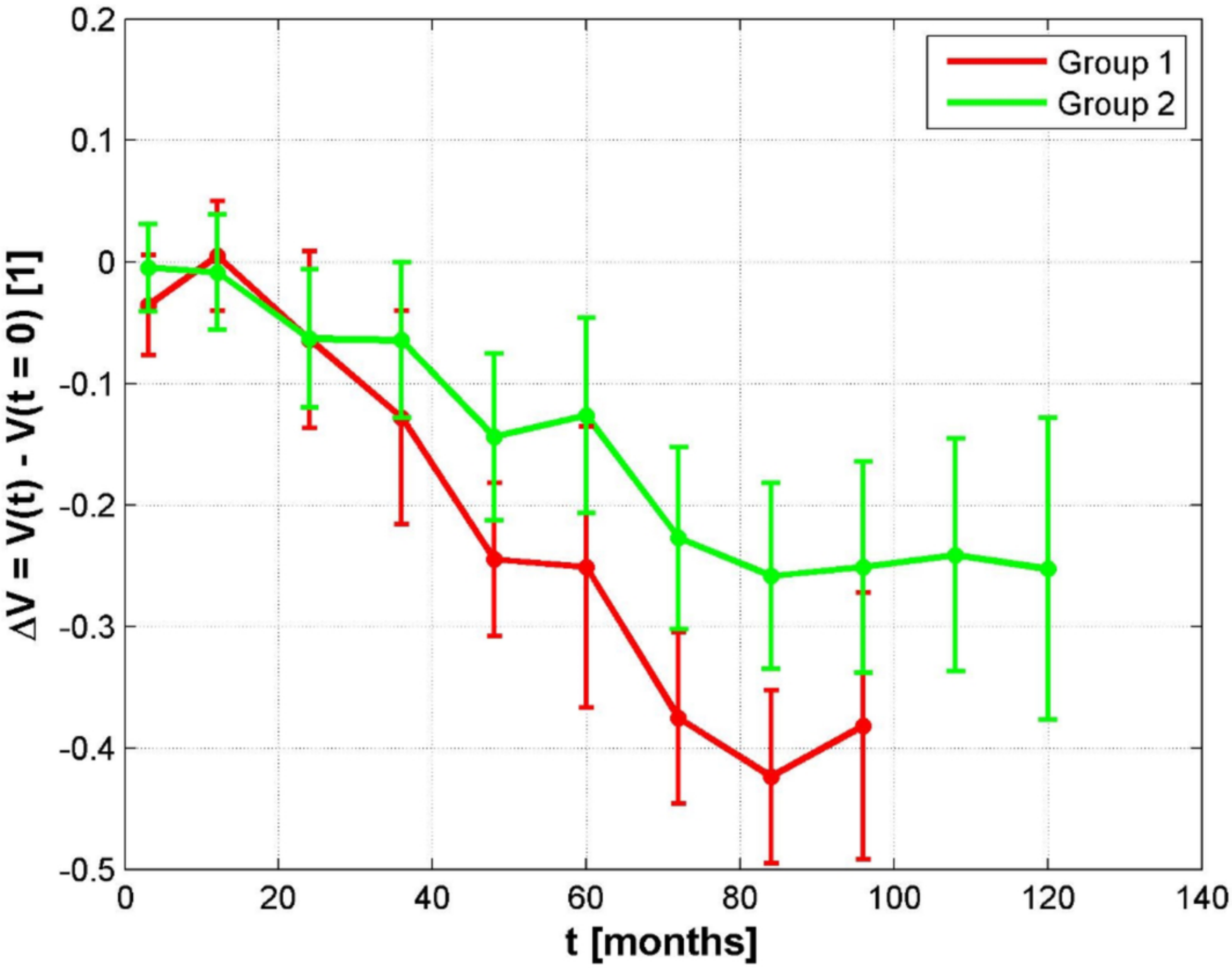
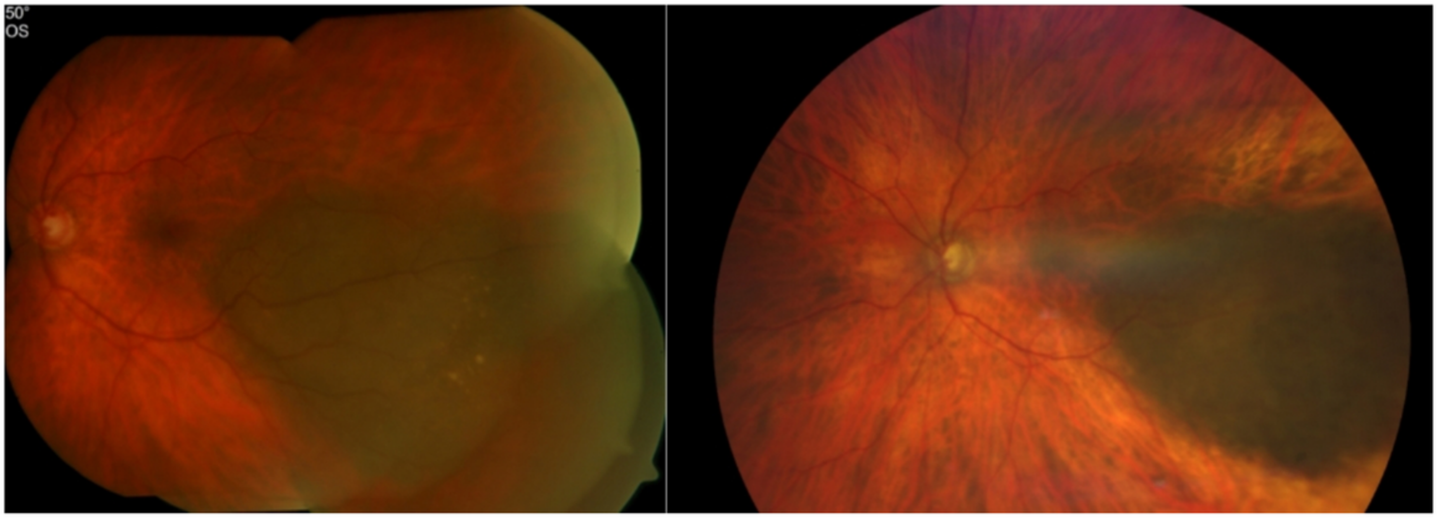


Figure 2

Change in visual acuity in groups I and II. The mean values are shown with the “standard error of the mean” dispersion measure.



A B

Figure 3

A case of patient with a CM in Group II: A: VA 1,0; Height 2,4mm. B: 90 months after brachytherapy, VA 0,8, Height 1,4mm